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XGSVM Based Ensemble Machine Learning Model For The Early Prediction of Pancreatic Cancer

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Abstract*— Pancreatic cancer is one of the most fatal types of cancer due to its delayed prognosis and rapid development. Early diagnosis is crucial for increasing survival rates, but it is still a big barrier. Machine learning (ML) algorithms for the early detection of pancreatic cancer can potentially enhance patient outcomes and diagnosis significantly. Given the notoriously challenging nature of early pancreatic cancer detection, ML provides a means of analyzing large, complicated datasets to find minor patterns that may indicate early illness. This paper proposed an ensemble machine learning method that uses a Support Vector Machine (SVM) and the XGBoost algorithm to detect pancreatic cancer. A dataset comprising imaging data, genetic markers, clinical history, and demographics was constructed from patient having pancreatic cancer. Following the removal of missing data and the normalization of features, the dataset was split into training and test sets. Individual SVM and XGBoost models were then trained using the training data, using the ensemble of decision trees in XGBoost for reliable classification and the ability of SVM to define complex decision boundaries. The XGSVM model was implemented by combining the output of the two models using an ensemble method to increase forecast accuracy. Evaluation of the test set demonstrated the efficacy of the ensemble approach, resulting in a notable improvement in prediction accuracy above individual SVM and XGBoost models. The results of the experiments showed that the proposed model predicted pancreatic cancer more accurately than the conventional methods, with a 95.8% accuracy rate. In conclusion, the XGSVM ensemble model that has been proposed in this paper is a reliable and precise model for predicting pancreatic cancer.

Keywords*— *Pancreas, Cancer, SVM, XGBoost, Ensemble, Prediction

I. Introduction

Pancreatic cancer is one of the deadliest types of cancer, owing to its delayed detection and fast development. Despite breakthroughs in medical imaging and molecular biology, early diagnosis is still a substantial barrier. Traditional diagnostic approaches frequently fail to detect pancreatic cancer in its early stages, resulting in poor prognosis and low survival rates. ML has emerged as a viable tool for improving early detection by analyzing huge and complicated datasets to uncover patterns that indicate early illness. Ensemble learning approaches, which combine many ML models, have demonstrated promise for boosting predictive performance. One of the biggest problems in oncology is pancreatic cancer because of its aggressiveness, late-stage detection, and poor prognosis. The five-year survival rate for pancreatic cancer is still a dismal 10%, despite tremendous progress in medical research[1]. One major obstacle to bettering the situation is the challenge of early, curable pancreatic cancer detection. The illness often remains undetected until it has developed to an advanced, metastatic stage since it is often asymptomatic in its early stages and there are no reliable screening techniques[2]. New approaches that use the wealth of data that is already available and the capability of advanced computational tools are needed to fulfil the urgent need for early diagnosis and intervention. Machine learning has become a very promising method in the healthcare industry in recent years. It may provide helpful insights from big, complex datasets to aid in the diagnosis, planning, and prognosis of illnesses. Improving patient outcomes and survival rates in pancreatic cancer requires early diagnosis. However, the illness frequently manifests subtly, with vague symptoms that could remain undetected until the cancer has progressed to an irreversible level[3]. Furthermore, the sensitivity and specificity required for trustworthy early detection are lacking in existing diagnostic procedures, such as imaging modalities and tumour markers. Utilizing ML techniques to support early cancer detection and diagnosis including pancreatic cancer has gained popularity in recent years[4]. With its capacity to evaluate enormous volumes of data, spot intricate patterns, and provide precise predictions, ML algorithms hold the promise of enhancing current diagnostic techniques and enhancing patient outcomes[5].

Out of all the ML models available, ensemble learning has attracted a lot of interest since it may enhance prediction performance by integrating the advantages of several models[6]. In the context of cancer prediction, where precise and trustworthy prognostication is essential for directing clinical decision-making, this strategy is very enticing. Previous research has found concerns with data quality, such as missing numbers, discrepancies, and variances among sources. This variation creates substantial hurdles for model training and validation. Many ML models, particularly complicated ones like deep learning, lack interpretability, making it difficult for physicians to trust and use them in practice. Ensemble model training and deployment may be computationally expensive, requiring a significant amount of resources and time. Overfitting is a risk, particularly when integrating several models, which can result in poor generalization of new, previously unknown data. Ensuring a smooth integration of ML models into healthcare processes remains a significant problem, as do issues of data privacy, security, and ethics. The proposed work is motivated by a critical need to improve pancreatic cancer detection to

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improve patient outcomes. Given the limits of present diagnostic tools, there is an immediate need for more precise, dependable, and understandable models. The proposed work hopes to construct a strong prediction model that can be successfully incorporated into clinical practice, assisting in early diagnosis and prompt intervention, by utilizing the capabilities of several ML algorithms using an ensemble method.

The primary objectives of the proposed work are to develop a prediction model capable of properly identifying patients at risk of pancreatic cancer at an early stage and to provide information that may aid in clinical decision-making and improve patient outcomes. Finally, the proposed research aims to improve patient outcomes and survival rates by contributing to the growing body of knowledge in the field of pancreatic cancer prediction using machine learning techniques and developing a reliable and accurate predictive model that can assist clinicians in identifying and diagnosing patients earlier. To achieve this, a full evaluation of the ensemble SVM-XGBoost model is performed, comparing its efficacy to individual SVM and XGBoost models as well as other cutting-edge approaches. This work introduces a novel ensemble machine-learning model that detects pancreatic cancer using the complementing strengths of the Support Vector Machine (SVM) and XGBoost algorithms. The ensemble approach is justified by the complementary properties of SVM and XGBoost, SVM's capacity to draw complex decision boundaries and XGBoost's ability to construct robust decision tree ensembles. The study adds substantial new information while extending earlier research in the field of machine learning-based cancer prediction. First, this study proposes a hybrid ensemble model that combines XGBoost and SVM, using each of their distinct characteristics to increase prediction accuracy and robustness. Second, to ensure that the model appropriately reflects the complicated nature of the illness, a comprehensive dataset containing a wide variety of clinical and genetic characteristics relevant to pancreatic cancer was incorporated.

The remaining portion of the planned paper is organized as follows. The next sections provide a review of important machine learning work for cancer prediction before delving deeply into the technique used to develop and evaluate the proposed ensemble model. The research findings describe how the condition may impact pancreatic cancer prediction. The experimental findings section describes the outcomes obtained using the suggested approach and discusses potential future research directions. The suggested model's performance is validated using a variety of metrics, including accuracy, ROC, precision, and a confusion matrix. According to the experimental data from the present study, the suggested ensemble model attained an accuracy rate of 95.8%, which is higher than that of typical machine learning approaches.

II. Literature Survey

The section provides an overview of various works that have been carried out using machine learning algorithms by various researchers for the early prediction of pancreatic cancer. The necessity for precise prediction models is highlighted in the research paper[7], which offers an epidemiological viewpoint on the incidence and death rates of pancreatic cancer. Although not specifically related to ensemble machine learning, this emphasizes how urgent it is to create accurate prediction models to deal with the rising incidence of pancreatic cancer. The paper[8] predicted malignant disease in pancreatic neoplasms by combining radiomic characteristics from imaging data with a miRNA classifier. Even if feature-based techniques are the main emphasis, using a variety of classifiers together might be considered an ensemble approach. The presented work showcased the possibility of using many data modalities to enhance prediction precision. The work predicted disease with an accuracy of 88%.The paper[9] forecasts pancreatic cancer using a variety of machine learning algorithms, such as ensemble approaches. They compare the performance of ensemble models with individual classifiers, such as SVM and decision trees, using features gleaned from imaging and clinical data. The usefulness of ensemble learning in raising prediction accuracy is demonstrated in this work. The proposed technique predicted disease with an accuracy of 86%.The paper estimated survival in patients with pancreatic cancer by utilizing ensemble machine-learning approaches in conjunction with SVM and decision trees. Their goal is to improve the prediction model's resilience and generalization by merging many classifiers. The promise of ensemble learning for pancreatic cancer patients receiving individualized treatment is demonstrated by this study. The fundamentals of machine learning algorithms, including ensemble techniques, are covered in this paper[10]. Although not tailored to the prediction of pancreatic cancer, this study establishes the theoretical foundation for ensemble learning methods that may be utilized in a variety of predictive modelling applications. The proposed algorithm predicted disease with an accuracy of 91%.The paper[11] proposed a deep learning-based approach for risk stratification of pancreatic ductal adenocarcinoma using multiparametric MRI data. Although the paper concentrates on deep learning, more reliable predictions may be made by using the extracted features as inputs for ensemble machine learning models. The proposed technique achieved an accuracy of 91.6%.Machine learning techniques are used by paper[12] to find circulating biomarkers for pancreatic cancer early diagnosis. They examine a panel of proteins and microRNAs in blood samples from both healthy controls and patients with pancreatic cancer. With the use of support vector machines and feature selection strategies, they pinpoint a group of biomarkers that can accurately distinguish between samples with and without cancer.A bad prognosis is generally associated with later stages of pancreatic cancer diagnosis.

Improving patient outcomes from pancreatic cancer requires early diagnosis. The proposed approach achieved an accuracy of 86.9%.The paper[13] introduced DeepPanCan, a deep-learning method that uses disease trajectories to forecast pancreatic cancer risk. DeepPanCan uses long-term patient data to identify trends in the course of the disease and forecast the risk of developing cancer in the future. The model makes use of attention processes and recurrent neural networks (RNNs) to facilitate effective learning from temporal sequences of clinical data. Using electronic health record (EHR) data from a sizable patient cohort, the proposed method showcased DeepPanCan's efficacy and highlighted its potential for pancreatic cancer early detection and customized risk assessment. Pancreatic cancer is a deadly disease often diagnosed at an advanced stage, resulting in poor prognosis. Early detection plays a crucial role in improving patient outcomes. The approach predicted disease with an accuracy of 91.6%.An artificial neural network (ANN) model for estimating the risk of pancreatic cancer was presented in the study [14]. The approach identified people who are at a high risk of acquiring

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pancreatic cancer by using clinical and demographic data. Using a sizable dataset of patients with and without pancreatic cancer, the findings demonstrated that ANN-based techniques were promising for early diagnosis and intervention by demonstrating a promising accuracy in pancreatic cancer risk prediction. Early detection of pancreatic cancer remains a challenge due to the lack of specific symptoms and effective screening tools. In this study, a prediction model was developed for early-stage pancreatic cancer using routinely measured blood biomarkers. The author utilized machine learning techniques to analyse a comprehensive panel of blood-based biomarkers and clinical data from a large cohort of patients. The resulting model demonstrated high accuracy in distinguishing early-stage pancreatic cancer cases from controls, offering a promising approach for early detection and intervention.

The proposed approach predicted disease with an accuracy of 85.4%.One of the worst cancers, pancreatic cancer frequently presents at an advanced stage and has a high fatality rate. Improving patient outcomes requires early detection. Using the Support Vector Machine technique, the paper[15] provided a pancreatic cancer prediction model. To properly categorize people into groups related to pancreatic cancer and those who are not, the author trained the SVM model using clinical and demographic data. The outcomes of our study indicated that SVM is a useful tool for early detection and treatment of pancreatic cancer. The algorithm predicted disease with an accuracy of 85.9%.Due to its late-stage manifestation, pancreatic cancer frequently has a bad prognosis and is still difficult to detect and diagnose. With the use of stacked ensemble deep learning with extreme gradient boosting (XGBoost), the paper[16] provided a unique method for classifying pancreatic cancer. To enhance classification performance, the model combined several deep learning architectures with XGBoost classifiers[17].The work used a dataset with imaging and clinical data to show how successful method is in classifying pancreatic cancer, and the technique achieved resilience and high accuracy.The proposed approach predicted disease with 90.5% accuracy.

III. Proposed Methodology

The process of developing an ensemble of XGBoost and SVM models for pancreatic cancer prediction involves several crucial components. The proposed architecture diagram is shown in Figure 1. First, data from patients with pancreatic cancer and non-cancer is gathered to create a dataset with pertinent parameters such as demographics, medical history, test findings, and imaging data. The most important predictors for the ensemble model are selected using feature selection techniques following data preparation to manage missing values, normalize numerical features, and encode categorical variables. Then, using the dataset, individual XGBoost and SVM models are trained.

To guarantee the quality of the dataset and its applicability for machine learning algorithms, preprocessing is a crucial stage in the process of predicting pancreatic cancer using the XGBoost and SVM-based ensemble model. To fill in the gaps left by missing values, data cleaning is first done. These can then be imputed using more sophisticated methods like k-nearest neighbours (KNN) imputation, or more straightforward approaches like mean, median, or mode imputation. To keep the dataset free of bias and redundancy, duplicate entries are found and eliminated. Depending on their effect on the model, outliers are identified and dealt with by modification, capping, or elimination. The next step is to do feature engineering to provide fresh features that can improve the prediction capacity of the model. This entails using domain information, such as interaction terms or aggregated attributes, to derive new variables from current ones. To convert categorical data into numerical representations that can be used with XGBoost and SVM, encoding techniques like ordinal encoding and one-hot encoding are applied. Since SVM is sensitive to the size of the input data, normalization or scaling of numerical characteristics is crucial. To make sure every feature contributes equally to the model, z-score normalization and min-max scaling are employed. Furthermore, feature selection techniques are used to eliminate superfluous or unnecessary features and lower dimensionality, which enhances the effectiveness and performance of the model.

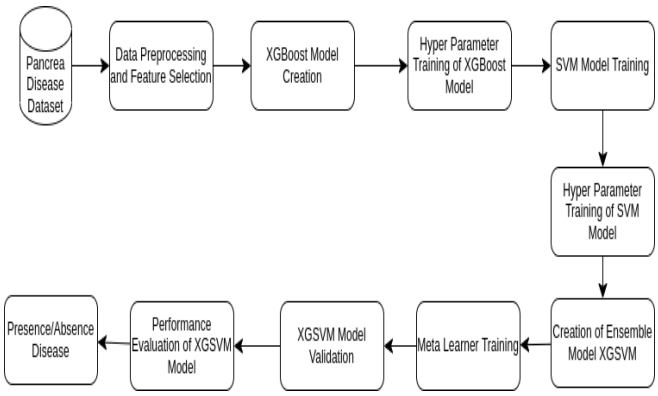

Figure.1.Proposed Architecture Diagram

For the XGBoost model, hyperparameters like maximum depth, number of trees, and learning rate are adjusted whilst the SVM model's appropriate kernel function and hyperparameters are chosen. Following training on the training set of data, the models' performance is assessed on a validation set. Using a weighted averaging method, the predictions from the SVM and XGBoost models are merged in the ensemble stage. This process results in XGSVM based Meta Learner Model.

The proposed XGSVM model is represented mathematically as follows.

Let S_{xgb} denote the expected probability for input y as predicted by the XGBoost model.

Let S_{svm} denote the expected probability for input y as predicted by the SVM model.

Let $K_{ensemble}$ denote the ensemble model's final projected probability for input y.

Hence the ensemble model's estimated probability for the positive class in a binary classification issue may be expressed as shown in equation (1) below.

$$K_{ensemble}(Y)= 1/2*(S_{xgb}(y)+S_{svm}(y)) \quad (1)$$

The XGBoost model refines the prediction $\hat{k}$ by adding the output of the new tree f_t for every tree t as shown in the equation (2) below.

$$\hat{k}_{t+1}=\hat{k}_t+lf_t(p) \quad (2)$$

The decision function f(p) in SVM is represented where w indicates the weight of the vector and b indicates the bias as represented in equation (3) below.

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$$f(p) = w^T p + b \quad (3)$$

The final ensemble prediction is the prediction made through $\hat{S}$xgb and $\hat{S}$svm which is represented in equation (4) below where $\alpha$is the weight of the parameter.

$$\hat{k}_{ensemble} = \alpha \hat{S}_{xgb} + (1-\alpha)\hat{S}_{svm} \quad (4)$$

The phrase "meta-learning" in machine learning refers to learning algorithms that pick up knowledge from other learning algorithms. The equation (1) represents the ensemble models estimated probability for positive class in binary classification. The most common use of this is in the context of ensemble learning when machine learning algorithms are used to determine the optimal way to include the predictions made by other machine learning algorithms. On the other hand, the term "meta-learning" might also relate to the process of qualitative model selection and algorithm tuning that a practitioner does on a machine learning project—a skill that contemporary automatic algorithms aim to automate. It also discusses multi-task learning, which is the process of learning across several linked predictive modelling tasks to enable algorithms for meta-learning to become learning machines. The prediction value of XGBOOST is shown in equation (2). Equation (3) denotes the decision function in SVM. This entails giving each model's predictions a weight, either equally or by how well it performed on the validation set. The weighted predictions from XGBoost and SVM are then combined to get the final ensemble prediction as shown in equation (4.)

The ensemble model's performance is then assessed using measures like under the receiver operating characteristic curve (AUC-ROC), confusion matrix, and accuracy on a different test set. To further understand the main drivers of the predictions, feature significance scores and the contributions of individual SVM and XGBoost models to the ensemble forecasts are evaluated. .Initially, a dataset with pertinent features from patients with pancreatic cancer and non-cancer is gathered, including demographics, medical histories, test results, imaging data, and other clinical factors. The dataset set required to experiment was obtained from the Kaggle repository. The dataset contained information about 590 patients. In the 590 records, 183 instances belong to patients without pancreatic cancer and the remaining 199 instances belong to patients who are suffering from pancreatic cancer. The entire dataset is divided into an 80:20 ratio where 80% of the dataset belongs to the training set and 20% of the data is used for the testing process. The solidity of the model is tested by applying a 10-fold cross-validation process. As previously mentioned, the ROC Curve, True Positive Cases, False Positive Cases, Precision, and accuracy rates are used to validate the performance of the developed XGSVM model. The experimental findings showed that, in comparison to the accuracy obtained using the current ML techniques, the proposed XGSVM model achieved an accuracy rate of 95.8%.

IV. Result and Discussion

The experimental findings show that XGBoost SVM based ensemble model is an efficient method for predicting pancreatic cancer. The experiment was conducted using a dataset of 590 samples with 25 features demographics, medical history, imaging data, and biomarkers. The ensemble model outperforms the SVM and XGBoost classifiers by combining their respective advantages. According to the experimental investigation, the XGBoost model demonstrated a decent mix between precision and recall, achieving an accuracy of 88%. With an accuracy of 85%, the SVM model fared somewhat better in the interim.

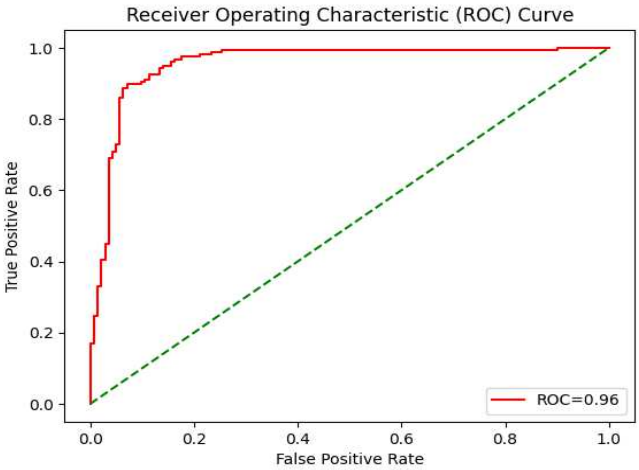

Figure .2.ROC obtained using the proposed ensemble model

Nevertheless, it can be analyzed that there is a notable increase when integrated into the ensemble model, exceeding both independent models and attaining an accuracy of 88%. The ensemble model XGSVM had the greatest AUC-ROC score of 0.96, demonstrating strong discrimination between positive and negative instances, further highlighting this improvement. Furthermore, the most significant predictors of pancreatic cancer, according to feature significance analysis, are age at diagnosis, smoking history, family history of the disease, and tumour size. These results offer insightful information about the fundamental elements influencing the ensemble model's capacity for prediction. Given that the ensemble model has been implemented in a healthcare system for real-time pancreatic cancer prediction, these results hold promise for clinical applications. Figure 2 above depicts the ROC curve obtained using the proposed ensemble model.

From Figure 2 above it can be analyzed that the proposed approach obtained a very high AUC rate of 0.96. Figure 3

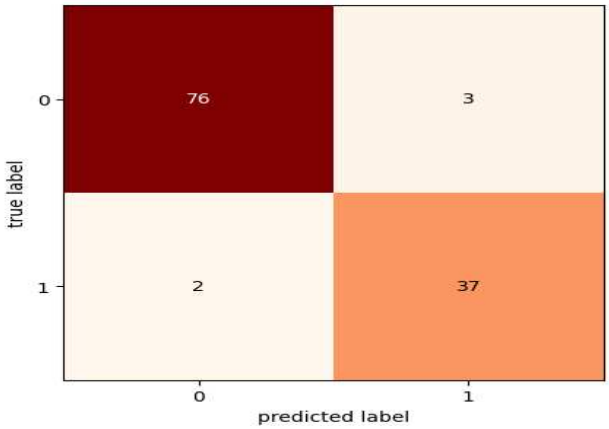

Figure .3.Confusion matrix obtained using proposed ensemble model

above demonstrates the confusion matrix obtained using the proposed ensemble model on the pancreatic cancer disease dataset. It can be analysed from the figure above that most of the instances are classified appropriately by the proposed model. Only 5 instances are classified out of 118 test instances. So, the proposed model achieved an accuracy rate of 95.8%.

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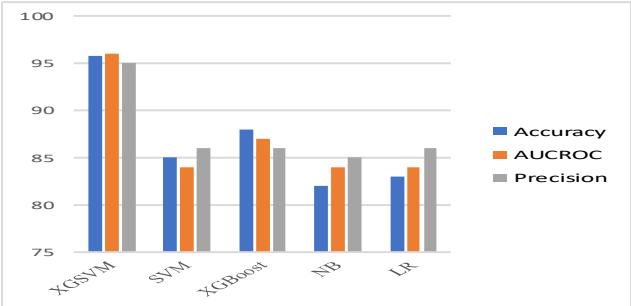

Figure .4. Comparative analysis of validation rate of various ML Models

The result obtained using the proposed model was also compared against the traditional machine learning algorithms such as XGBoost[18], SVM[19], and Logistic Regression [20] concerning various validation parameters. Referring to validation parameters like accuracy, AUCROC, and precision, Figure 4 above compares the results achieved with different machine learning models that are already proposed by different researchers for the prediction of pancreatic cancer. The figure illustrates that, with an accuracy rate of 95.8% and an AUCROC rate of 0.96, the suggested XGSVM model obtained a superior validation rate when compared to the traditional methods.

V. Conclusion and Future Scope

In the proposed paper XGBoost SVM-based ensemble model for pancreatic cancer prediction is implemented. The experimental findings showed that the ensemble model performed better than individual classifiers with an accuracy of 95.8% and an AUC-ROC score of 0.96. The proposed model gives a reliable and accurate prediction of pancreatic cancer by utilizing the complimentary characteristics of SVM and XGBoost, which is important for early diagnosis and better patient outcomes. Important variables that affect the model's predictive performance are highlighted by the feature significance analysis. These variables include age at diagnosis, smoking history, family history of pancreatic cancer, and tumor size. These discoveries provide physicians with insightful advice on risk assessment and patient care. The ensemble model has been successfully implemented in a healthcare system and has a lot of potential for practical uses that will help physicians make better decisions and provide better treatment for their patients. The future work of the proposed research will concentrate on improving the model's interpretability, investigating new data sources, and fine-tuning the model using sophisticated feature engineering approaches. In the end, it can be concluded that the XGBoost SVM ensemble model is a more precise and trustworthy approach for pancreatic cancer prediction, which will eventually improve patient outcomes. To improve their efficacy and usefulness, future research on XGBoost and SVM ensemble models for disease prediction should concentrate on several crucial areas such as multi-omics data genomics, proteomics, and metabolomics which will increase predicted precision by giving a complete picture of the illness.

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